

Cooperative Transportation of a Cable-Suspended Load by Multiple Quadrotors^{*}

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Abstract: This paper presents the problem of cooperative transportation of cable-suspended load by four quadrotors, considering lifting and carrying it to the desired pose. Quadrotors and the load in the system, are modeled using Newton-Euler approach. The load is modeled assuming it as a point load, which means that the cable goes from the quadrotors CoM (Center of Mass) to the load CoM. There is switching in dynamics when load exerts a force in the drones in comparison when quadrotors are moving free. Under these considerations, the system can be modeled using a hybrid systems approach, bearing in mind that the system is switching the dynamics when quadrotors start to lift the load. Quadrotors make use of geometric control to track the desired trajectory, which is created by implementing way-points in an optimization process. In this sense, a position-based passivity formation control is implemented in order to create a formation capable of reducing efforts. Reducing the effort made by quadrotors when carrying a common load, is also achieved by adding a new term in the control law in charge of keeping drones in a desired formation even when their position is perturbed by the load. Finally, improvements in trajectory tracking and the reduction of position errors are shown through simulation.

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1. INTRODUCTION

In recent years, aerial robotic platforms such as quadrotors have been widely explored and implemented by academic and governmental research groups worldwide. Quadrotors are characterized by having well maneuverability and mobility in a highly-constrained three-dimensional environment. Quadrotors have been considered as a useful robot in a great variety of applications such as surveillance, aerial construction, exploration of unknown areas, and SAR (Search and Rescue) tasks (Yanguas-Rojas et al. (2017); Sousa et al. (2019); and Cardona and Calderon (2019)). In SAR missions, quadrotors are used as explorers since not only are they able to scan and map the affected area looking for victims in less time than humans but also, because they can avoid putting more lives at risk. It is worth to mention that, the greater number of quadrotors the more accurate and robust the mission. In view of, the failure of one robot does not affect the fulfillment of the entire mission. Moreover, another task that quadrotors could make in a disaster area is the deployment of the first aid kit, as shown in Bernard et al. (2011) and by Loianno and Kumar (2018). Considering that they can move by air being faster than humans do, due to time is critical at the moment of save lives in a disaster zone.

Despite the great feasibility of using drones to transport cargo, quadrotors are usually limited in terms of their payload carrying capacity. Nevertheless, the problem of

a single quadrotor carrying a load has been studied considering minimal load swing. Oscillation minimization is achieved by the creation of optimal trajectories based on either dynamical programming, as shown in Palunko et al. (2012) or reinforcement learning approach, as shown in Faust et al. (2013). On the other hand, a lifting maneuver of a payload by a single quadrotor using a hybrid approach to model the system is depicted in Cruz and Fierro (2017). For their purposes, they consider hybrid modes such as setup, pull, and raise that represents the dynamics of the operating moment. However, quadrotors can carry payloads beyond their limits by collaborating and performing cooperative transportation. Where the dynamics of the system is exposed in Sreenath and Kumar (2013) and Gassner et al. (2017). Furthermore, control performed by them to accomplish the task is a differential hybrid flatness. On the contrary, in Lee et al. (2010) it is shown the modeling process of a quadrotor and a geometric control applied with the aim of tracking a position and attitude reference. In which outputs are the thrust needed in each motor.

In this paper, we consider a load that is lifted by a group of robots attached from their CoM (Center of Mass) to the point-mass of the load. Modeling the system through Newton-Euler approach, bearing in mind a hybrid model which is caused by the switching dynamics, as shown in Cruz and Fierro (2014) and Cruz and Fierro (2017) but considering that there is more than a single robot. According to this, the main contribution of this work is twofold. First, we consider a multi-robot system attached

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to a common load while minimizing the effort generated in each robot by the load. This objective is achieved with the use of a position-based control using passivity approach whose output is the desired relative positions with respect to the load. Second, the use of geometric control in a local way in each robot with the purpose of tracking the desired trajectory, but with a global objective that lifts a load while minimizing the effort in a cooperative way. The rest of the document is organized as follows: Section II shows the problem statement in which the dynamics of each state of the system is defined, Section III shows the design of the control law developed to take the system to the desired behavior, Section IV presents the simulation case and some results, and finally Section V presents the document conclusions and some future work alternatives.

Notation and Definitions: $\{X_w, Y_w, Z_w\}$ unit vectors along the axes of $\{W\}$, $\{X_{B_i}, Y_{B_i}, Z_{B_i}\}$ unit vectors along the axes of the i^{th} quadrotor $\{B_i\}$ with respect to $\{W\}$, $m_{q_i} \in \mathbb{R}_{>0}$ mass of the i^{th} quadrotor, $J_i \in \mathbb{R}^{3 \times 3}$ inertia matrix of the i^{th} quadrotor with respect to $\{B_i\}$, $r_{q_i}, v_{q_i} \in \mathbb{R}^3$ position and velocity of the i^{th} quadrotor center-of-mass in $\{W\}$, $R_i \in \mathbb{R}^3$ the rotation matrix from $\{B_i\}$ to $\{W\}$, $\Omega_i \in \mathbb{R}^3$ angular velocity of i^{th} quadrotor in $\{B_i\}$, $F_i \in \mathbb{R}$ total thrust produced by the i^{th} quadrotor $M_i \in \mathbb{R}^3$ moment produced by the i^{th} quadrotor, $m_l \in \mathbb{R}_{>0}$ mass of the load, $r_l, v_l \in \mathbb{R}^3$ position and velocity of the load in $\{W\}$, $l_i \in \mathbb{R}_{>0}$ length of the cable from load to i^{th} quadrotor, $T_i \in \mathbb{R}_{\geq 0}$ tension on the cable produced by the i^{th} quadrotor and the load.

2. PROBLEM STATEMENT

Consider a group of N aerial robots that navigate in a convex bounded environment $S \in \mathbb{R}^3$. Which means that robots are capable to calculate the distance from one point to another one using the Euclidean distance. In this paper, the way robots communicate among them is modeled by the use of a graph $G = (V, E)$ which is a representation of the system that is composed of nodes and links, where $V = \{v_1, v_2, \dots, v_i\}$ represents the quadrotors and $E = \{(v_1, v_2), \dots, (v_1, v_j), (v_2, v_j), \dots, (v_i, v_j)\}$ communication or interaction links. A communication link captures the capability of each robot to have access to the information of others. A given assumption is the SLAM (Simultaneous Localization and Mapping) process of each robot, which means that robots can determine the pose of the load and those robots which it has communication with.

Multi-quadrotor Dynamics. Applying Newton-Euler approach is possible to model the motion of the system as a summation of forces and torques. Obtaining in this way the model used for a group of N robots in this work $N = 1, 2, 3, 4$ drones without being attached to a load as follows

$$\begin{cases} \dot{r}_{q_i} = v_{q_i}, \\ m_{q_i} \dot{v}_{q_i} = m_{q_i} g(-Z_w) + F_i(Z_{B_i}), \\ \dot{R}_i = R_i \dot{\Omega}_i, \\ J_i \dot{\Omega}_i = -\Omega_i \times J_i \Omega_i + M_i, \end{cases} \quad (1)$$

where state variables are defined in Notation and Definitions (Section 1) $Z_{B_i} = R_i Z_w$ in which $Z_w = u_3$ with $u_3 = [0 \ 0 \ 1]^T$, g is the constant gravitational acceleration

applied in Z_w and the hat map $\hat{\cdot}: \mathbb{R}^3 \rightarrow SO(3)$ represents the skew-symmetric operator matrix as explained in Mahony et al. (2012), we define for instance $w_i = [w_1, w_2, w_3]$ then

$$\hat{W} = \begin{pmatrix} 0 & -w_3 & w_2 \\ w_3 & 0 & -w_1 \\ -w_2 & w_1 & 0 \end{pmatrix}. \quad (2)$$

Multi-quadrotor Dynamics Lifting a Load. Following the same procedure described previously, it is possible to identify the changes that the system has at the moment when the quadrotors start the lifting maneuver. We obtain the dynamics of the system as follows

$$\begin{cases} \dot{r}_{q_i} = v_{q_i}, \\ m_{q_i} \dot{v}_{q_i} = m_{q_i} g(-Z_w) + F_i R_i(Z_w) + T_i(\mu_i), \\ \dot{R}_i = R_i \dot{\Omega}_i, \\ J_i \dot{\Omega}_i = -\Omega_i \times J_i \Omega_i + M_i, \\ \dot{r}_l = v_l, \\ m_l \dot{v}_l = -\sum T_i(\mu_i) - m_l g(Z_w), \end{cases} \quad (3)$$

where μ_i is the unit vector that goes from the load towards the i^{th} quadrotor, considering also that the position of the load r_l can be determined by the following expression

$$r_l = r_{q_i} - l_i \mu_i.$$

Cooperative Transport Approach. Quadrotors i and j transporting an object exert forces F_i, F_j (for robots attached at the i^{th} and j^{th} contact points, we assume always i and j positionally opposed to each other). The vectors to these contact positions (in the body fixed frame) are expressed as μ_i, μ_j as shown in Fig. 1. These forces are summed (over N robots) to give the net force on the object. Let γ be the vector of net external forces on the body, then for the planar case (altitude remaining as constant), $\gamma_x = \sum_{i=1}^N F_{i,x}$, $\gamma_y = \sum_{i=1}^N F_{i,y}$. In this kind of problem is shown in Kumar and Waldron (1988) that if there is no interaction force then the following equation holds

$$(F_i - F_j) \cdot (\mu_i - \mu_j) = 0, \quad (4)$$

this is called the Zero Interaction Force Condition (ZIF) which is one of the control objective.

3. CONTROL DESIGN

In order to accomplish the desired task, which is lifting a suspended-cable load with the use of four quadrotors and take it to a desired position while satisfying (4), the tracking trajectory control is designed. Firstly, it is necessary to create an optimal reference trajectory that take the quadrotor from one initial position to a final desired position. This trajectory generator considers the quadrotor dynamics. In the case of the z-axis, $s = [s_1 \ s_2 \ s_3]^T = [z_q \ \dot{z}_q \ \ddot{z}_q]^T$ the dynamic system can be defined as, $\dot{s}_1 = s_2$, $\dot{s}_2 = s_3$, and $\dot{s}_3 = u$ with $s_0 = [0 \ 0 \ z_0]^T$, the desired trajectory is found by minimizing the following cost function

$$L = \alpha(s_{1_f} - z_f)^2 + \beta s_{2_f}^2 + \int_0^{t_f} u^2 dt, \quad (5)$$

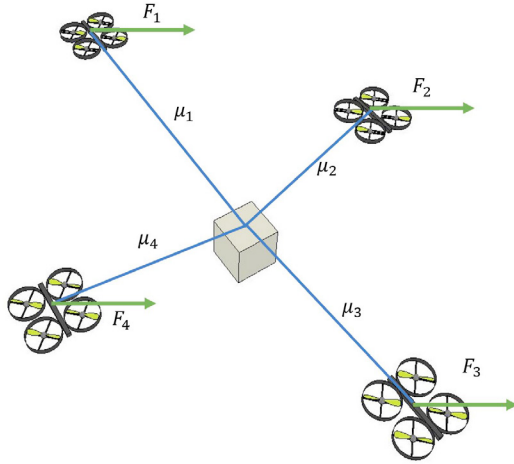


Fig. 1. Cooperative transportation: carrying a load by the cooperation of a group of quadrotors.

where z_0 is the initial position, z_f is the desired final position, t_f is the time defined to go from z_0 to z_f , s_{1_f} and s_{2_f} are the values of s_1 and s_2 at final time t_f and α, β are positive constants. Once that the function cost is solved the trajectory generator always will need either an initial and final position each time as needed. Taking this into account, a formation position control based in passivity is proposed. This formation control guarantee that no matter the initial position of the drones are, they will make the desired shape that minimizes the effort generated by the load and the other quadrotors.

3.1 Position Based Passivity Controller

As foregoing mentioned this controller works with the purpose to find the desired position at the end of the first sub-trajectory that place the robots in good positions to lift the load. First, a matrix that captures the interaction dynamics of the system is defined, called incidence matrix, which is filled as follows

$$d_{ik}(G) := \begin{cases} 1 & \text{if } k \in L_i^+ \\ -1 & \text{if } k \in L_i^- \\ 0 & \text{otherwise} \end{cases},$$

where L_i^+ and L_i^- represents if the link k^{th} points in or out of node i^{th} respectively. Additionally, the dynamic model employed for the target positions that will be delivered to the trajectory generator are a double integrator model. The double integrator dynamics takes the form

$$u_i = m_i \ddot{v}_{q_i}, \quad i = 1, \dots, N, \quad (6)$$

where u_i is the force input control, m_i is the mass of agent i^{th} and $\dot{v}_{q_i}$ is the velocity of agent i^{th} . When the control signal is added, the dynamic equation can be expressed as

$$u_i = -k_i(v_{q_i} - v_r) + m_i \dot{v}_r + u_i^d, \quad k_i \geq 0, \quad (7)$$

where v_r is the common and reference velocity, k_i is a proportional controller. We consider a group of agents modeled like the equation (6), the control law that is implemented to obtain the function in (7) seeks to guarantee that $(\dot{r}_{q_i} - v_r) \rightarrow 0$ and $\Delta_k \rightarrow \Delta_k^d$. According to Corollary 2.1 in Bai et al. (2011) the global asymptotic stability of the system is guaranteed by

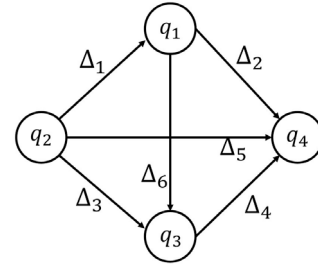


Fig. 2. Digraph scheme.

$$\dot{r}_{q_i} = H_i \left(-\sum_{k=1}^N d_{ik} \psi_k(\Delta_k) \right) + v_r, \quad i = 1, \dots, N,$$

where H_i denotes the output at time t of a static or dynamic block satisfying $|\dot{r}_{q_i} - v_r| \rightarrow 0$ and $(r_{q_i} - r_{q_j}) \rightarrow 0$ as $t \rightarrow \infty$ for every pair of nodes (v_i, v_j) which are connected by a path. Δ_k denotes the differences between r_{q_i} and its neighbors r_{q_j} . $\psi_k: \mathbb{R}^p \rightarrow \mathbb{R}^p$ are non-linearities designed such that the target set where the dynamics evolves is invariant and asymptotically stable. In this work the target set can be defined as $\Lambda = \{\Delta_k \mid \Delta_k = \Delta_k^d\}$ where Δ_k^d dictates the desired relative configuration of the nodes. The control law then is

$$u_i^d = -\sum_{k=1}^N d_{ik} \psi_k(\Delta_k - \Delta_k^d),$$

giving as a result of replacing in (7) as

$$\begin{aligned} m_i(\ddot{r}_{q_i} - \dot{v}_r) + k_i(\dot{r}_{q_i} - v_r) \\ + \sum_{k=1}^N d_{ik} \psi_k(\Delta_k - \Delta_k^d) = 0, \end{aligned} \quad (8)$$

the consensus protocol starts from a potential quadratic function taking into account $\psi_k(\Delta_k) = \nabla P_k(\Delta_k)$ and is related to (8) as

$$P_k = \frac{\delta_k}{2} \|\Delta_k - \Delta_k^d\|^2, \quad \delta_k \in \mathbb{R}_{>0}$$

$$\psi_k(\Delta_k) = \delta_k(\Delta_k - \Delta_k^d),$$

the δ_k constants are the system feedback gains which regulate the difference between the desired set and current system conditions. According to our requirements, Δ_k is chosen in the form that the desired shape is a rhombus. Where quadrotors 1 and 3 are aligned and positional opposed in relation to the load and quadrotors 2 and 4 as well as shown in Fig. 2.

3.2 Geometric Control F_i

As a result of (1), the variables of control are two. First, F_i that corresponds to the total thrust produced by the four motors in the i^{th} quadrotor. Second, the moments executed in each axis M_i that will be shown in the next sub-section. As mentioned in Section 1, geometric control is implemented as depicted in Lee et al. (2010). The procedure begins by defining the position and velocity errors as $e_{p_i} = r_{q_i} - r_i^{des}$ and $e_{v_i} = v_{q_i} - \dot{r}_i^{des}$ respectively, in this way the desired thrust takes the form as follows

$$\begin{aligned} F_i^{des} = & -K_p e_{p_i} - K_v e_{v_i} + m_{v_i}(g u_3 + \ddot{r}_i^{des}) \\ & -K_g((F_i - F_j) \cdot (\mu_i - \mu_j)), \end{aligned} \quad (9)$$

where $m_{v_i} = m_{q_i} + m_l$, K_p and K_v are diagonal gain matrices and $\|F_i^{des}\| \neq 0$ at any time, but the traslational

motion of the quadrotor is controlled in the projection of F_i^{des} on to the third body-axis giving as a result

$$F_i = F_i^{des} \cdot Z_{B_i}.$$

3.3 Geometric Control M_i

On the other hand, the variable responsible to control the altitude error and the angular velocity is M_i . In the case of the altitude error, that stabilize the translational movement can be chosen as

$$Z_{B_i}^{des} = \frac{F_i^{des}}{\|F_i^{des}\|},$$

taking F_i^{des} from (9). Here the yaw angle of the quadrotor remains in zero, allowing to choose $X_{B_i}^{des} = [1 \ 0 \ 0]^T$, obtaining a desired pose as

$$R_i^{des} = [Y_{B_i}^{des} \times Z_{B_i}^{des} \ Y_{B_i}^{des} \ Z_{B_i}^{des}] \in SO(3),$$

where

$$Y_{B_i}^{des} = \frac{Z_{B_i}^{des} \times X_{B_i}^{des}}{\|Z_{B_i}^{des} \times X_{B_i}^{des}\|},$$

and finally defining the pose error as

$$e_{R_i} = \frac{1}{2} ((R_i^{des})^T R_i - R_i^T R_i^{des})^\vee, \quad (10)$$

where $\vee$ represents the vee operator which is the inverse of the skew symmetric operator defined from (2) as

$$w^\vee = \begin{pmatrix} -W_{23} \\ W_{13} \\ -W_{12} \end{pmatrix}.$$

As foregoing mentioned under the assumption to remain yaw angle as constant the angular velocity can be defined as was made in Mellinger and Kumar (2011), $\Omega_i^{des} = \rho_i^{des} X_{B_i}^{des} + \varrho_i^{des} Y_{B_i}^{des}$, where $\rho_i^{des} = -h_\omega \cdot Y_{B_i}^{des}$ and $\varrho_i^{des} = h_\omega \cdot X_{B_i}^{des}$ where

$$h_\omega = \frac{m_w}{F_i} (r_i^{des} - (Z_{B_i}^{des} \cdot r_i^{des}) Z_{B_i}^{des}),$$

thus the angular velocity error is

$$e_{\Omega_i} = \Omega - \Omega_i^{des}, \quad (11)$$

then in the same way as in the geometric control of F_i the control of M_i taking (10) and (11) is as follows

$$M_i = -K_R e_{R_i} - K_\Omega e_{\Omega_i} + (\Omega_i \times J_i \Omega_i),$$

where again K_R and K_Ω are diagonal gain matrices that guarantee the stability of the system.

4. SIMULATION AND RESULTS

In order to demonstrate the improvements implemented, two simulations are carried out allowing us to compare the position errors in the quadrotors. First, we contemplate the switching dynamics but without the fourth term in (9), which is the geometric control developed in Lee et al. (2010). This algorithm needs predefined waypoints as inputs for the geometric control whose is in charge of taking the quadrotor to the desired position. Second, we perform the simulation that gathers all the concepts mentioned in the previous section in which we show the

Table 1. Simulation Parameters

Parameter	Value	Parameter	Value
$r_{q1}(0)$	$[-0.75, 0.5, 0]$	$r_{q2}(0)$	$[0.75, 0.5, 0]$
$r_{q3}(0)$	$[0.75, -0.5, 0]$	$r_{q4}(0)$	$[0.75, -0.5, 0]$
$v_{q1}(0)$	$[0, 0, 0]$	$v_{q2}(0)$	$[0, 0, 0]$
$v_{q3}(0)$	$[0, 0, 0]$	$r_{q4}(0)$	$[0, 0, 0]$
J_1	$I_3[11.7, 11.7, 23.4]$	m_{q1}	0.505
J_2	$I_3[11.7, 11.7, 23.4]$	m_{q2}	0.505
J_3	$I_3[11.7, 11.7, 23.4]$	m_{q3}	0.505
J_4	$I_3[11.7, 11.7, 23.4]$	m_{q4}	0.505
l_1	1.5	l_2	1.5
l_3	1.5	l_4	1.5
k_p	$I_3[1.58, 1.56, 1.73]$	t_f	10
k_v	$I_3[0.79, 0.79, 0.71]$	α	10
k_R	$I_3[11, 11.7, 7.3]$	β	10
k_Ω	$I_3[1.9, 1.7, 0.29]$	m_l	0.1
k_g	$I_3[0.1, 0.1, 0]$	δ_i	I_3
$v(t)$	0	k_i	5

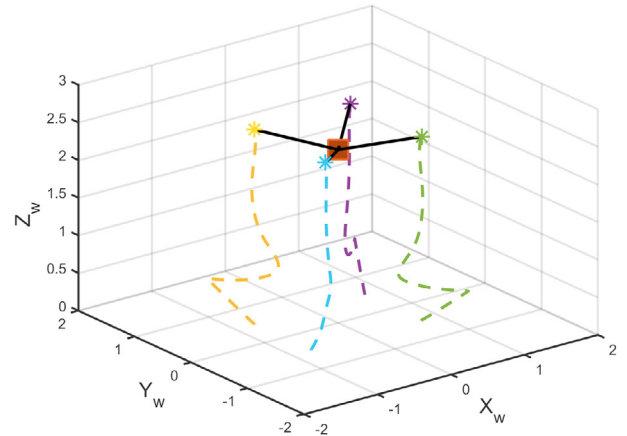


Fig. 3. Cooperative transportation: Simulation using conventional geometric control.

improvements in the position error in comparison with the one obtained in Lee et al. (2010). The algorithm is capable to take random initial quadrotors positions as long as they are physically possible. This means that robots cannot be able to be far from the load since they are attached to the load. This is possible by applying the position-based passivity controller that takes the initial position of each quadrotor and gives the first group of waypoints. Those waypoints are the ones capable of minimizing lifting efforts satisfying (4). In this particular case when using four quadrotors the formation is a rhombus, due to in this shape quadrotors minimize the efforts. Once the first desired position is achieved, the dynamics of the system change to the one that contemplates the mass in the model. In the same way, to lift the load to the desired altitude the control must be modified considering this time the fourth term in (9) and bearing in mind that the mass of the body is changing due to the load weight. Table 1 defines the simulation parameters needed.

To start analyzing the results, the two simulations previously mentioned are shown in a 3-D plot in Fig. 3, and Fig. 4, where dashed lines correspond to the trajectory performed by each quadrotor. Asterisks represent

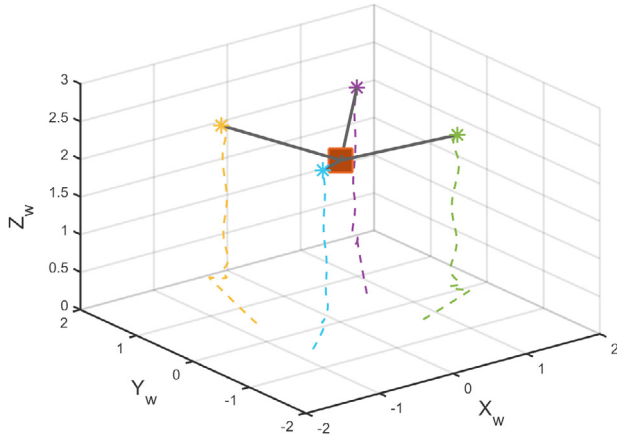


Fig. 4. Cooperative transportation: Simulation considering improvements of positioning and minimization of effort.

the quadrotors, and the solid dark gray lines are the cables attached to the brown square load. As expected, regardless of the controller implemented, both are capable to take the load to the desired altitude. However, there are some changes, on the one hand, the first simulation which does not contemplate the improvements presents translations changes in x and y -axis suddenly, and in an abrupt way. On the other hand, the second simulation also presents changes in x and y -axis, but they are shorter and smoother. The changes in x and y -axis are due to the new force added when the cable is stretched to the limit and the load starts to influence it. On the other hand, the position error is reduced by the improvements applied in the second simulation. It is worth to mention that in the single quadrotor, lifting the load the tension can be found by summation of forces in the entire system, giving as a result

$$T = \frac{m_l}{m_q + m_l} F,$$

considering that the forces are all of them in the z -axis because the quadrotor is exactly over the load. In contrast, in the multi-quadrotor system, the effects of the forces depend on unit vectors μ_i and also the summation of forces changes to

$$\frac{T_1 + T_2 + T_3 + T_4}{m_l} = \frac{F_1 - T_1}{m_{q_1}} + \frac{F_2 - T_2}{m_{q_2}} + \frac{F_3 - T_3}{m_{q_3}} + \frac{F_4 - T_4}{m_{q_4}},$$

this is something expected, taking into account (3), where the dynamic of the mass depends on the summation of all the tensions generated in each quadrotor. What this new term in the control shows is that placing the robots aligned with its positional opposed quadrotor improves the performance of the task.

As mentioned before, the x and y -axis are the most affected by the use of multiple robots to lift a suspended-cable load. The reason is that the forces are not just in the z -axis, since the vectors going from quadrotors to mass

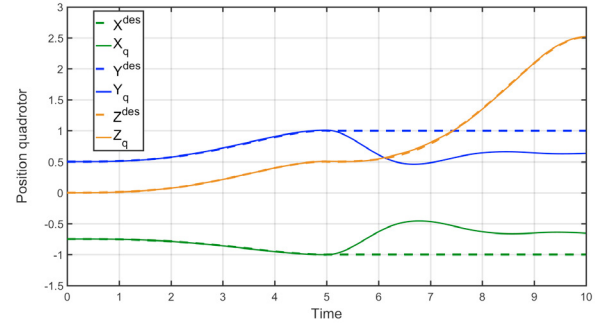


Fig. 5. Cooperative transportation: Position evolution in x, y , and z using conventional geometric control.

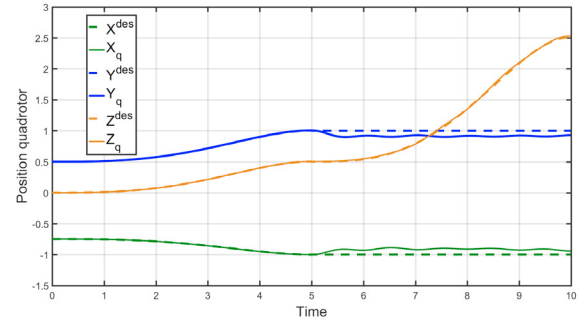


Fig. 6. Cooperative transportation: Position evolution in x, y and z considering improvements of positioning and minimization of effort.

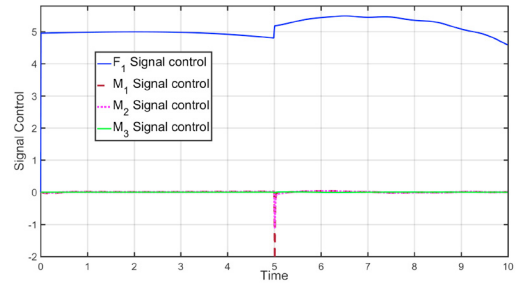


Fig. 7. Cooperative transportation: Control Signals, using conventional geometric control.

μ_i have a force in each component. Fig. 5 and Fig. 6 show the evolution of each axis over time. The z -axis is represented by orange lines, x -axis with blue lines and y -axis with green lines. In the same way, dashed lines are the desired trajectory generated by solve (5), and solid lines are the evolution of the r_{q_1} states. As shown in the second simulation, considering the improvements is way better than in the first simulation, where just the conventional geometric control is implemented. Regarding z -axis, both satisfy the task but, just the modified controller can make the position error tend to zero. In the second simulation quadrotors can correct the position in x and y faster than in the first simulation. This could be thanks to the modification in the control law as this term tries to maintain the alignment between the opposed quadrotors, thus minimizing the efforts and adding and additional force to these axes in the controller.

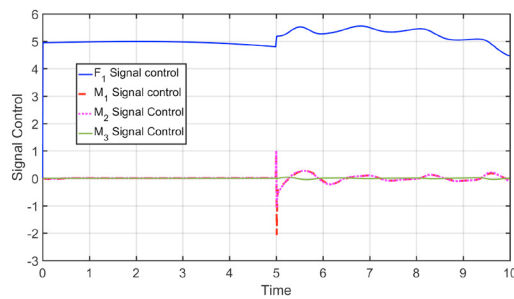


Fig. 8. Cooperative transportation: Control Signals, considering improvements of positioning and minimization of effort.

Finally, Fig. 7 and Fig. 8 show the signal control applied to quadrotor 1 in both cases, first, without additional term in the geometric control and second with it, respectively. The blue solid line represents the F output control, the green solid line is the first term of vector M in the geometric control, the dashed magenta line is the second term and dashed red line is the third one. As can be seen, F signal changed in the second simulation with some oscillations in the control caused by the additional term. In the same way, vector M presents oscillations in the second and third terms, which are the signal produced to correct the position in a faster way and avoid the trajectory to separate in the initial lifting effort.

5. CONCLUSION AND FUTURE WORK

As shown in the previous section, the implementation of geometric control to track the desired trajectory in multiple-drones was reached. When attaching a common load to multiple quadrotors, the dynamics of the entire system changes but still the regular geometric control is capable of lift the load. However, it fails to keep a formation and track the desired position in x and y -axes. The reason is the tension generated by the load in each quadrotor, considering also that the more quadrotors exert forces in different directions, the more disturbance each robot feel. Looking for the ZIF condition and adding it to the control law shows a better performance in the x and y -axes and the quadrotors were more synchronized. On the other hand, position-based passivity controller demonstrates that regardless of the initial position, quadrotors always find the desired shape. Being the one that minimizes the effort in each quadrotor, considering the tension on each cable attached to the load.

For future work, it is proposed to include disturbances to the dynamics associated with wind, uncertainties, and changes in trajectory in one of the quadrotors. It is also worth to develop a control that changes the shape and distribution of forces when one or more of the quadrotors either stop working or fails. Additionally, a machine learning algorithm can be applied to find better gains control and improve the performance. Finally, the navigation and cooperative transportation can be implemented in a non-convex scenario while maintaining the formation shape.

REFERENCES

Bai, H., Arcak, M., and Wen, J. (2011). *Cooperative control design: a systematic, passivity-based approach*.

- Springer Science & Business Media.
- Bernard, M., Kondak, K., Maza, I., and Ollero, A. (2011). Autonomous transportation and deployment with aerial robots for search and rescue missions. *Journal of Field Robotics*, 28(6), 914–931.
- Cardona, G.A. and Calderon, J.M. (2019). Robot swarm navigation and victim detection using rendezvous consensus in search and rescue operations. *Applied Sciences*, 9(8), 1702.
- Cruz, P. and Fierro, R. (2014). Autonomous lift of a cable-suspended load by an unmanned aerial robot. In *proc. 2014 IEEE Conference on Control Applications (CCA)*, 802–807.
- Cruz, P.J. and Fierro, R. (2017). Cable-suspended load lifting by a quadrotor uav: hybrid model, trajectory generation, and control. *Autonomous Robots*, 41(8), 1629–1643.
- Faust, A., Palunko, I., Cruz, P., Fierro, R., and Tapia, L. (2013). Learning swing-free trajectories for uavs with a suspended load. In *proc. 2013 IEEE International Conference on Robotics and Automation*, 4902–4909.
- Gassner, M., Cieslewski, T., and Scaramuzza, D. (2017). Dynamic collaboration without communication: Vision-based cable-suspended load transport with two quadrotors. In *proc. 2017 IEEE International Conference on Robotics and Automation (ICRA)*, 5196–5202.
- Kumar, V.R. and Waldron, K.J. (1988). Force distribution in closed kinematic chains. *IEEE Journal on Robotics and Automation*, 4(6), 657–664.
- Lee, T., Leok, M., and McClamroch, N.H. (2010). Geometric tracking control of a quadrotor uav on se (3). In *proc. 49th IEEE conference on decision and control (CDC)*, 5420–5425.
- Loianno, G. and Kumar, V. (2018). Cooperative transportation using small quadrotors using monocular vision and inertial sensing. *IEEE Robotics and Automation Letters*, 3(2), 680–687.
- Mahony, R., Kumar, V., and Corke, P. (2012). Multirotor aerial vehicles: Modeling, estimation, and control of quadrotor. *IEEE robotics & automation magazine*, 19(3), 20–32.
- Mellinger, D. and Kumar, V. (2011). Minimum snap trajectory generation and control for quadrotors. In *proc. 2011 IEEE International Conference on Robotics and Automation*, 2520–2525.
- Palunko, I., Cruz, P., and Fierro, R. (2012). Agile load transportation: Safe and efficient load manipulation with aerial robots. *IEEE robotics & automation magazine*, 19(3), 69–79.
- Sousa, P., Ferreira, A., Moreira, M., Santos, T., Martins, A., Dias, A., Almeida, J., and Silva, E. (2019). Isep/inesc tec aerial robotics team for search and rescue operations at the eurathlon 2015. *Journal of Intelligent & Robotic Systems*, 93(3-4), 447–460.
- Sreenath, K. and Kumar, V. (2013). Dynamics, control and planning for cooperative manipulation of payloads suspended by cables from multiple quadrotor robots. *rm*, 1(r2), r3.
- Yanguas-Rojas, D., Cardona, G.A., Ramirez-Rugeles, J., and Mojica-Nava, E. (2017). Victims search, identification, and evacuation with heterogeneous robot networks for search and rescue. In *proc. 2017 IEEE 3rd Colombian Conference on Automatic Control (CCAC)*, 1–6.